

YAML

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What is YAML?

- YAML is a Data Serialization Language.
- A New Language for Data
- Modeled after Perl-like structures
- But not just intended for Perl
- Human friendly, cross language, Unicode based data serialization language.

- YAML can be taken as an acronym for either
 - **Yet Another Markup Language**
 - **YAML Ain't Markup Language**
- Like JSON, the purpose of YAML is to represent typical data types in human-readable notation

Why YAML if XML?

- Unlike XML which is not easily readable by humans, YAML was created to be **human-friendly** and integrate easily with modern programming languages.
- Unlike with XML, YAML was intended to simplify the viewing and understanding of **config files, log files, object persistence, and messaging**, to allow the programmer to spend more time programming and less time formatting data.

YAML Design Goals

YAML design goals:

- - YAML documents are very readable by humans.
- - YAML interacts well with scripting languages.
- - YAML uses host languages native data structures.
- - YAML has a consistent information model.
- - YAML enables stream-based processing.
- - YAML is expressive and extensible.
- - YAML is easy to implement.

Design consideration

- To achieve Human/Computer easy processing
- High Data/Markup (Signal/Noise) Ratio
- To be the world's best Data::Dumper
- To be great for Configs, Caches, and Logs
- Unicode based
- Speedy parsing
- Alternative to XML (for some applications)

Basic YAML Syntax Rules

- Documents begin with `---` and end with `...`
- Indentation of lines denotes the structure within the document.
- Comments begin with `#`
- Members of lists begin with `–`
- Key value pairs use the following syntax `<key>:value>`
- YAML files should end in `.yaml`
- YAML is case sensitive.
- YAML does not allow the use of tabs. Spaces are used instead as tabs are not universally supported.

YAML Syntax Basics

- Collections
- Streams and Documents
- Mappings (hashes / dictionaries)
- Sequences (arrays / lists)
- Comments

Contd..

- Scalars (strings,integers, dates and other atomic data types)
 - Simple
 - Quoted
 - Block (|)
 - Folded
 - Wiki
 - Escaping
- Anchors & Aliases (&,*)
- Tags (Data types)(“!”)
- Nodes -YAML [nodes](#) have content of one of three kinds: scalar, sequence, or mapping. In addition, each node has a [tag](#) which serves to restrict the set of possible values which the node's content can have.

Collections:

- YAML's [block collections](#) use [indentation](#) for scope and begin each entry on its own line.

Example 2.1. Sequence of Scalars
(ball players)

```
- Mark McGwire  
- Sammy Sosa  
- Ken Griffey
```

Example 2.4. Sequence of Mappings
(players' statistics)

```
-  
  name: Mark McGwire  
  hr:   65  
  avg:  0.278  
-  
  name: Sammy Sosa  
  hr:   63  
  avg:  0.288
```

YAML uses three dashes (“`---`”) to separate documents within a stream. Three dots (“`...`”) indicate the end of a document without starting a new one

Example 2.7. Two Documents in a Stream
(each with a leading comment)

```
# Ranking of 1998 home runs
---
- Mark McGwire
- Sammy Sosa
- Ken Griffey

# Team ranking
---
- Chicago Cubs
- St Louis Cardinals
```

Example 2.8. Play by Play Feed
from a Game

```
---
time: 20:03:20
player: Sammy Sosa
action: strike (miss)
...
---
time: 20:03:47
player: Sammy Sosa
action: grand slam
...
```

- A question mark and space (“? ”) indicate a complex [mapping key](#).
- & and * indicate Anchors and Aliases

Example 2.10. Node for “Sammy Sosa”
appears twice in this document

```
---  
hr:  
  - Mark McGwire  
  # Following node labeled SS  
  - &SS Sammy Sosa  
rbi:  
  - *SS # Subsequent occurrence  
  - Ken Griffey
```

Example 2.11. Mapping between Sequences

```
? - Detroit Tigers  
  - Chicago cubs  
:  
  - 2001-07-23  
  
? [ New York Yankees,  
    Atlanta Braves ]  
: [ 2001-07-02, 2001-08-12,  
    2001-08-14 ]
```

Mappings

- A YAML mapping is like a Perl hash
- Unordered Key/Value pairs
- Separated by ': ' (space is mandatory)

name: Benjamin

rank: Private

serial number: 1234567890

12:34 PM: My favorite time

Sequences

- A YAML sequence is like a Perl array
- An ordered collection of data
- YAML has a bullet like syntax '- '

- red
- white
- blue
- pinko

Examples

Fruits:

- Apples
- Tomatoes

Veggies:

- Spinach
- Broccoli

Meats:

- Burgers
- Shrimp

Household:

- Candles
- Incense
- Toilet Duck

Ex2:The Matrix

-

- 3

- 5

- 7

-

- 0

- 0

- 7

-

- 9

- 1

- 1

Comments

- Comments/blank lines can go almost anywhere
- Must not be ambiguous with content
- Comments begin '#'

```
# comment before document
--- #DIRECTIVE # comment
foo: bar # inline comment
phone: number #555-1234
    ### Comment
fact: fiction
---
blue: bird
# Comment
```

Scalars

- A YAML scalar is strings, integers, dates and other atomic data types.
- YAML has many scalar forms.

```
---
```

```
simple: look ma, no quotes
```

```
quoted:
```

- 'Single quoted. Like Perl, no escapes'
- "Double quotes.\nLike Perl, has escapes"
- |

```
  A YAML block scalar.
```

```
  Much like Perl's
```

```
  here-document.
```

Scalar content can be written in block form using a literal style (“|”) where all line breaks count. Or they can be written with the folded style (“>”) where each line break is folded to a space unless it ends an empty or a “more indented” line.

Example 2.16. Indentation determines scope

```
name: Mark McGwire
accomplishment: >
  Mark set a major league
  home run record in 1998.
stats: |
  65 Home Runs
  0.278 Batting Average
```

Tags

- <http://yaml.org/type/>

- YAML Pros:
 - Cleaner format
 - Safe deserialization
 - Hashes sorted by default
 - Supports GLOB(*.txt), Regexp
 - Interacts with Ruby, Python, Java etc

--- !<tag:clarkevans.com,2002:invoice>

invoice: 34843

date : 2001-01-23

bill-to: &id001

given : Chris

family : Dumars

address:

lines: |

458 Walkman Dr.

Suite #292

city : Royal Oak

state : MI

postal : 48046

ship-to: *id001

product:

- sku : BL394D

quantity : 4

description : Basketball

price : 450.00

- sku : BL4438H

quantity : 1

description : Super Hoop

price : 2392.00

tax : 251.42

total: 4443.52

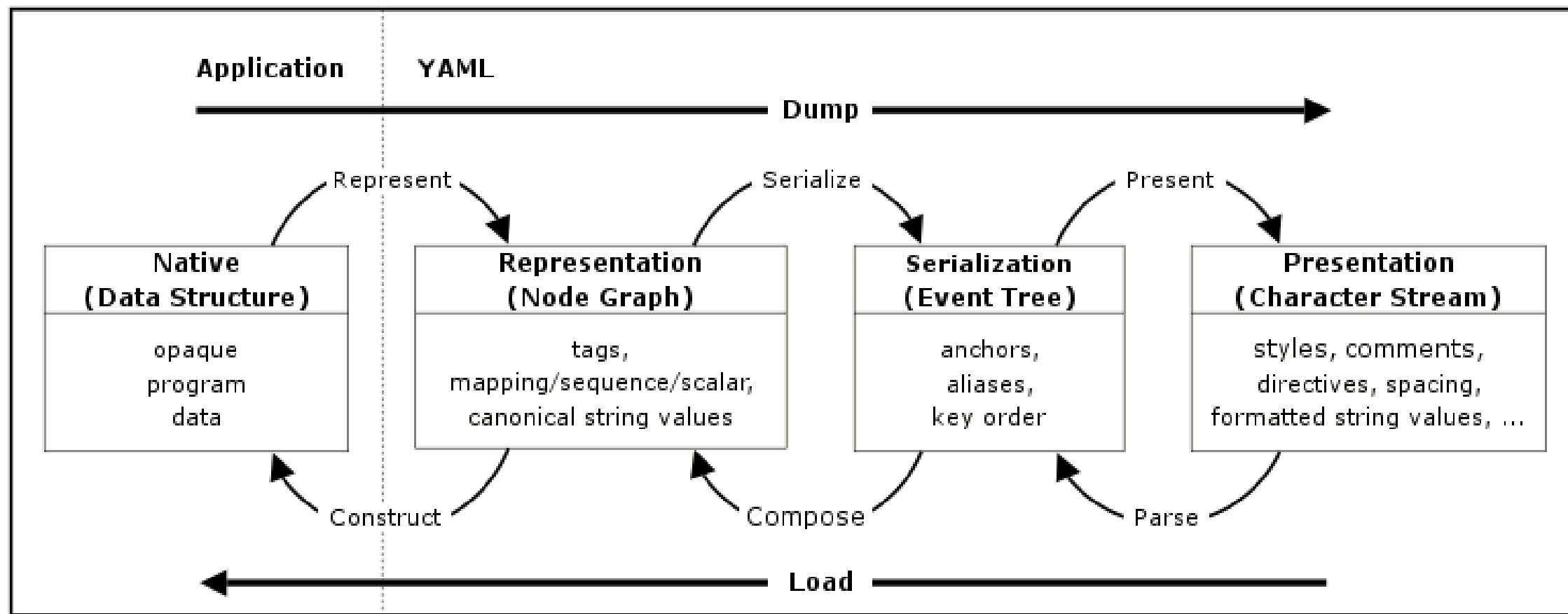
comments:

Late afternoon is best.

Backup contact is Nancy

Billsmer @ 338-4338.

```
---
Time: 2001-11-23 15:01:42 -5
User: ed
Warning:
  This is an error message
  for the log file
---
Time: 2001-11-23 15:02:31 -5
User: ed
Warning:
  A slightly different error
  message.
---
Date: 2001-11-23 15:03:17 -5
User: ed
Fatal:
  Unknown variable "bar"
Stack:
- file: TopClass.py
  line: 23
  code: |
      x = MoreObject("345\n")
- file: MoreClass.py
  line: 58
  code: |-
      foo = bar
```



Demo

<https://gsabarinath91@bitbucket.org/gsabarinath91/serialisation.git>

References:

- <https://learnxinyminutes.com/docs/yaml/>
- <http://pyyaml.org/wiki/PyYAMLDocumentation>
- <http://yaml.org/refcard.html>
- <http://sangsoonam.github.io/2017/03/13/yaml-vs-json.html>